

C1Q/SPP1 Macrophage Spatial Context as a Traceability Feature in Non-Small Cell Lung Cancer: A Public Multi-Cohort Cancer-Informatics Study

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Abstract

Objectives: Public cancer-informatics datasets can test whether immune programs detected in bulk tumor data remain traceable across spatial, genetic, and clinical contexts. We asked whether C1Q/SPP1 is a macrophage-context traceability feature in non-small cell lung cancer (NSCLC). The objective was evidence synthesis and hypothesis prioritization, not clinical classifier development, immunotherapy selection, causal proof, or therapeutic targeting.

Methods: We analyzed TCGA NSCLC for discovery, feature stability, and internal score-behavior assessment, and GSE31210 for external prognostic traceability among patients with complete follow-up. Two small NSCLC immune-checkpoint-blockade (ICB) cohorts and one melanoma ICB cohort were analyzed separately as exploratory response-association datasets. Spatial context was assessed using 10x Visium and NanoString CosMx SMI data, with interpretation anchored to patient, section, or sample-contrast units rather than spot or cell counts. Mendelian-randomization analyses were used only for expression-linked gene prioritization. Harmonized PD-L1, tumor mutational burden, and treatment metadata were unavailable across cohorts; incremental clinical utility was not claimed.

Results: Nested cross-validation identified a stable 91-feature set enriched for immune and macrophage-context signals. C1QA, C1QB, and C1QC showed section-level spatial autocorrelation in patient-limited spatial datasets (Moran's I approximately 0.35-0.39), whereas SPP1 showed stronger spatial clustering (Moran's I = 0.712). Retrospective algorithm benchmarking showed detectable but modest prognostic score behavior in TCGA and attenuated external traceability in GSE31210 (AUC = 0.650; C-index = 0.629). ICB cohorts showed directional response-associated patterns, but small NSCLC sample sizes and inclusion of a melanoma exploratory cohort precluded validation-level claims.

Conclusions: C1Q/SPP1 is best interpreted as a traceable macrophage-spatial context in public NSCLC resources and a candidate axis for follow-up study. The findings are hypothesis-generating, not a validated clinical biomarker, treatment-selection rule, causal mechanism, or therapeutic-target claim. Prospective, NSCLC-specific, spatially resolved evaluation is required before clinical implementation can be considered.

Keywords: Non-small cell lung cancer; Cancer informatics; C1Q; SPP1; macrophage context; spatial transcriptomics; reproducibility; traceability feature

Introduction

Non-small cell lung cancer (NSCLC) remains a major cause of cancer mortality.¹⁻³ Immune checkpoint blockade has improved outcomes for a subset of patients.^{4,5} However, PD-L1 expression, tumor mutational burden, and bulk immune-expression summaries do not fully capture the spatial and cellular organization of the tumor microenvironment (TME).^{8,9} A central cancer-informatics problem is therefore not simply to build another retrospective classifier, but to determine whether candidate immune signals remain interpretable when traced across independent data types. Spatial transcriptomics and single-cell technologies now make it possible to examine whether immune programs visible in bulk data also map to spatial contexts that can be traced across available datasets.¹⁴⁻¹⁶

C1q, encoded by *CIQA*, *CIQB*, and *CIQC*, is produced largely by macrophage and dendritic-cell lineages in the TME. Prior work suggests that complement-associated macrophage states may accompany either immune activation or tumor-promoting inflammation, depending on context. *SPP1*-positive macrophage programs, by contrast, are often linked to stromal remodeling and immune exclusion. These observations motivate a narrower and more testable question: can the C1Q/SPP1 macrophage axis be traced across public NSCLC transcriptomic, spatial, and clinical resources without overstating clinical or mechanistic certainty?

The present study is intentionally framed as a public cancer-informatics evidence synthesis. Here, a traceability feature denotes a biologically interpretable signal that can be followed across public datasets and analytic layers without being promoted to a clinical assay. The term is an interpretation boundary, not a substitute for validation: it does not imply a locked diagnostic test, treatment-selection rule, clinical-use algorithm, or definitive mechanism. Instead, we evaluate whether C1Q/SPP1 links macrophage context, spatial organization, retrospective survival associations, and exploratory immunotherapy-response signals. This conservative framing separates data-supported observations from hypotheses that require prospective evaluation and experimental testing.

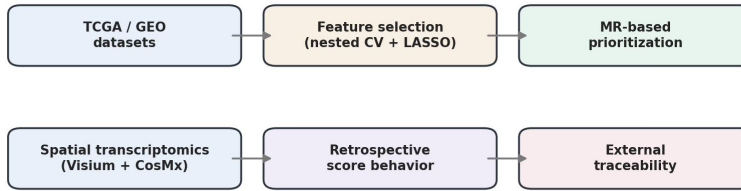
Methods

Study Design and Evidence Hierarchy

This study used a staged, retrospective public-data design.²⁵⁻²⁷ The evidence hierarchy was defined before interpretation: (1) TCGA NSCLC was used for discovery, feature stability, and internal score-behavior assessment; (2) GEO cohorts were used for external prognostic traceability where complete follow-up was available; (3) Visium and CosMx SMI datasets were used to evaluate spatial context, with inference emphasized at the patient, section, or paired-contrast level; and (4) immunotherapy-treated cohorts were treated as exploratory response-association datasets because of small sample sizes and tumor-type heterogeneity (Figure 1). No external cohort was used to tune the feature set, select a clinical threshold, or upgrade the analysis to prospective validation.

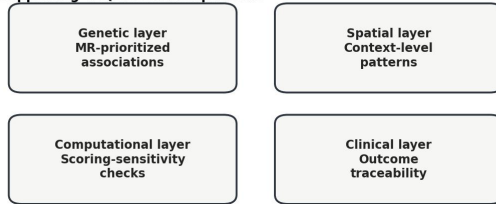
Integrated Study Design and Evidence Overview in NSCLC

Panel A Study design and analytical workflow



Workflow summary only; this panel describes the analytical sequence rather than a biological pathway.

Panel B Multi-layer evidence supporting C1Q-associated patterns



The four components are shown in parallel to indicate complementary evidence rather than a directed sequence.

Panel C Conceptual representation of spatial patterns

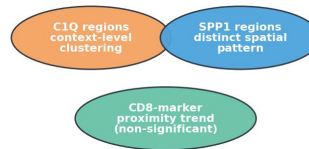


Figure 1. Integrated study design and evidence overview. Panel A, Analytical workflow: TCGA/GEO data acquisition, feature selection, MR-based prioritization, spatial transcriptomics, retrospective score-behavior assessment, and external traceability. Panel B, Multi-layer evidence framework: genetic, spatial, computational, and clinical evidence layers are interpreted as complementary support rather than a directed causal sequence. Panel C, Conceptual spatial-context representation: C1Q-enriched regions, SPPI-enriched regions, and CD8-marker context are shown as a schematic guide to the spatial analyses.

Conservative Interpretation Rules

The final interpretation followed five reporting rules. First, external survival cohorts were used to assess traceability of score behavior, not to claim clinical validation. Second, ICB-treated cohorts were analyzed as response-association context only, because the NSCLC cohorts were small and the melanoma cohort was not disease matched. Third, spot- and cell-level spatial measurements were treated as within-sample observations; patient, section, or paired-sample contrasts were the main interpretive units. Fourth, MR was used as an expression-linked prioritization layer under instrumental-variable assumptions and was not treated as proof of spatial mechanism, therapeutic actionability, or treatment response. Fifth, because PD-L1, TMB, ctDNA, TCR, and treatment metadata were not harmonized across the public cohorts, the study does not claim incremental clinical utility over standard biomarkers, stage-only models, clinical-only models, or published immune signatures.

Data Acquisition and Quality Control

Discovery Cohort (TCGA): TCGA LUAD and LUSC transcriptomic and clinical data were retrieved from the NCI Genomic Data Commons.^{28,29} A total of 1,100 TCGA NSCLC tumor samples were initially considered (595 LUAD and 505 LUSC). After quality control and availability filtering, 1,038 samples

contributed to discovery analyses. Analyses requiring complete survival, stage, or model features used the corresponding complete-case subsets; denominators are reported with each analysis.

Spatial Transcriptomics Data: For spatially resolved context analysis, we incorporated 10x Genomics Visium and Nanostring CosMx SMI datasets. Quality control, normalization, and platform-specific processing are described in the Supplementary Methods and source-data manifest. Spatial inference was deliberately anchored to patient, section, or sample-contrast units; spots and cells were treated as measurement units within those samples rather than as independent biological replicates.

10x Visium: Included 8 tissue sections from 1 patient (Supplementary Table S12), utilizing a 55 μm spot diameter for tissue-level spatial mapping.

CosMx SMI: Encompassed 8 tissue sections from 5 NSCLC patients, containing approximately 800,000 profiled cells targeted with a 960-gene RNA panel. Cell counts were treated as descriptive observations; independent inference was anchored to samples, sections, and paired contrasts.

External Cohorts: Clinical and expression data were retrieved from GEO. GSE31210 contained 226 lung adenocarcinoma samples after exclusion of non-tumor controls and 178 patients with complete follow-up for survival analysis. GSE31210 was used as an external traceability cohort and not as a locked prospective validation set. GSE126044 and GSE135222 were treated as small NSCLC ICB response-association cohorts, whereas GSE91061 was treated as a melanoma exploratory cohort only. These datasets were not pooled as confirmatory NSCLC immunotherapy evidence and were not used for algorithm selection or threshold optimization.

C1Q/SPP1 Traceability Framework

We implemented a multi-stage traceability framework to evaluate whether C1Q/SPP1 macrophage context is traceable across data layers:

Phase 1: MR-based gene prioritization: Mendelian randomization (MR) was used to prioritize genes with expression-linked immune-context associations.³³ It was not used to prove mechanism or treatment relevance. We used eQTL data from eQTLGen.³⁵ GTEx v8 and TwoSampleMR (v0.5.6) were used for supporting analyses, and sensitivity diagnostics were inspected before assigning confidence.

Phase 2: Retrospective score-behavior assessment: Machine-learning algorithms were benchmarked to assess whether the stable feature set carried outcome-linked signal. Apparent training-set metrics were retained only for audit transparency; interpretation focused on internal cross-validation, held-out score behavior, and external traceability. The resulting models were not interpreted as clinical tools and no fixed clinical cutoff is proposed.

Phase 3: Spatial context characterization: A C1Q/SPP1-centered scoring approach was used to evaluate macrophage-associated spatial organization and CD8-marker context in Visium and CosMx data. Spatial outputs were interpreted as context summaries rather than direct cell-cell signaling, causal recruitment, or patient-level spatial-classifier evidence.

These interpretation boundaries are summarized in Table 1.

Table 1. Claim-interpretation-boundary framework used for interpretation

Claim domain	Supported interpretation	Boundary
C1Q/SPP1 macrophage context	Traceable across public expression datasets as a macrophage-context feature	Not a diagnostic assay or tumor-specific marker
Spatial organization	C1Q and SPP1 show non-random spatial patterns in available Visium and CosMx sections	Spots/cells are measurement units, not independent biological replicates; patient-level extrapolation remains limited
MR-based prioritization	Highlights candidates for follow-up under instrumental-variable assumptions	Not causal proof, target validation, or evidence of therapeutic actionability
Retrospective outcome score	The feature set carries modest outcome-linked signal and can be used descriptively	No locked clinical cutoff, no prospective test, and no treatment-selection claim
ICB response association	Directional response-associated separation is observed in small cohorts	Not NSCLC ICB confirmation and not evidence of superiority over PD-L1, TMB, or clinical baselines

Note: This table defines the interpretive boundaries used throughout the manuscript and the public evidence-chain map.

Neighborhood Enrichment Analysis

Neighborhood enrichment analysis was performed in CosMx SMI data to describe local co-occurrence between C1Q-associated macrophage states and CD8-marker-positive contexts. A 50 μm radius was used around each index cell, and observed neighborhood composition was compared with a permutation-based null distribution (N=1,000 permutations). Because the number of independent patients was limited, these results were interpreted as local spatial-context support rather than definitive evidence of cell-cell signaling or causal recruitment.

Data Preprocessing and Statistical Analysis

Transcriptome Data Preprocessing

Bulk transcriptomic datasets were processed using dataset-appropriate normalization. TCGA RNA-seq expression values were filtered for low expression, log-transformed where appropriate, and aligned by gene symbol. External microarray data were quantile normalized or used as normalized series-matrix data when provided by GEO. Cross-platform analyses used within-cohort z-score scaling after feature alignment; platform-specific effects were not interpreted biologically. Excluded samples and feature-retention decisions are documented in the source-data manifest.

Statistical Analysis

All statistical analyses were performed using R and Python. Python-based data manipulation and numerical computing used Pandas, NumPy, and SciPy.³⁷⁻³⁹ Visualization used Matplotlib and Seaborn.^{40,41} Machine-learning tasks used Scikit-learn.⁴² Survival analysis used Kaplan-Meier curves and Cox proportional hazards models. Model performance was evaluated with time-dependent AUC, concordance index (C-index), and hazard ratios with 95% confidence intervals where available. P-values computed from spot- or cell-level spatial summaries were treated as within-section descriptive statistics, not as independent patient-level inference. Data and code availability are detailed in the Statements and Declarations section.

Sample Size and Statistical Power

The TCGA discovery cohort (N=1,038 after quality control) was adequate for exploratory feature discovery and internal score-behavior assessment. External cohorts were smaller and had lower event counts; therefore, external survival and immunotherapy-response results were interpreted as traceability or exploratory association rather than confirmatory evidence. No prospective sample-size calculation was possible because the study used public retrospective data. This limitation was not addressed by post hoc relabeling of external cohorts as validation sets.

Nested Cross-Validation Framework

To reduce data leakage and estimate feature stability, we used 5×5 nested cross-validation. The outer loop used stratified 5-fold splits for held-out assessment, and the inner loop used LASSO feature selection with cross-validated regularization. The final $\lambda = 0.002$ yielded 91 non-zero coefficients, and features selected in all outer folds were retained as the stability-prioritized set (Supplementary Table S4). This strategy separates feature selection from held-out fold assessment within TCGA but does not replace prospective testing. GSE31210 and the ICB cohorts were not used for feature selection, hyperparameter tuning, or cutoff determination.

Mendelian Randomization for Gene Prioritization

We used two-sample Mendelian randomization as a gene-prioritization layer.⁴⁴ MR was implemented in the TwoSampleMR R package. MR estimates were interpreted as expression-linked associations under instrumental-variable assumptions, not as proof of spatial mechanism, therapeutic actionability, or clinical utility.

Instrumental variables were selected at genome-wide significance ($P < 5 \times 10^{-8}$), LD-pruned ($r^2 < 0.001$, 10,000-kb window), and screened for instrument strength. IVW was used as the primary estimator, with MR-Egger, weighted median, MR-PRESSO, heterogeneity, pleiotropy, directionality, and leave-one-out analyses used as diagnostics. Exposures were eQTL-mediated gene-expression proxies; outcomes were immune-context or immune-infiltration proxies and should not be interpreted as directly measured single-cell densities or spatial phenotypes. Detailed diagnostics are provided in Supplementary Table S11 and Supplementary Figures S9–S10.

Spatial Transcriptomics Analysis

Spatial analyses were designed to test whether C1Q/SPP1-associated expression showed non-random local structure in the available tissue sections. Because the spatial datasets contained few independent patients, all spatial statistics were interpreted as descriptive spatial-context summaries.

Spatial Autocorrelation Analysis

$> P < 0.001$ We quantified spatial clustering of gene expression using Moran's I statistic implemented in the 'spdep' R package, a measure of spatial autocorrelation ranging from -1 (dispersion) to +1 (clustering). We calculated Moran's I values for candidate genes across spatial locations using a queen contiguity matrix with 10,000 permutations. Genes with Moran's I > 0.35 and corrected $P < 0.001$ were described as spatially clustered within the analyzed sections. The term hotspot is used descriptively and does not imply causal organization.

Cell Type Deconvolution

Cell-type context in Visium data was estimated using the GigaTIME framework and cross-checked against canonical marker expression, including CD68 for macrophage context and CD8A/CD8B for CD8-marker context. These outputs were used as contextual estimates rather than direct ground-truth cell counts. CosMx SMI was used to inspect single-cell-resolution expression and neighborhood patterns in the available sections.

Spatial Co-localization Analysis

To assess spatial context among C1Q, SPP1, and immune-marker programs, we calculated correlations and local spatial statistics across multiple spatial scales (50 μm , 100 μm , and 200 μm where available).^{47,48} Bivariate Moran's I, Geary's C, and local indicators of spatial association were used as descriptive spatial summaries with permutation-based significance testing and Benjamini-Hochberg correction. These analyses were interpreted as evidence of co-localization or spatial separation, not as proof of direct biological interaction.

Counterfactual Scoring and Sensitivity Analysis

We performed a counterfactual scoring analysis centered on *CIQA*. *CIQA* abundance was reduced computationally and the C1Q composite score, CD8 proxy, Moran's I, network summaries, and risk-score outputs were recomputed. This procedure tests score dependency and internal coherence of the C1Q module; it is not an experimental perturbation and is not interpreted as intervention-like evidence. Dose-response, mediation-style decomposition, bidirectional MR, Steiger directionality comparison, spatial scale-dependency, and specificity analyses were used as sensitivity analyses to assess whether the C1Q/SPP1 signal was internally consistent across analytic views.

Retrospective Score Construction and Evaluation

We benchmarked eleven candidate supervised algorithms or model families using the 91-feature set. Hyperparameters were optimized with Optuna-based Bayesian optimization and stratified cross-validation; detailed search spaces and selected parameters are listed in Supplementary Table S1. The purpose was to assess whether the feature set carried outcome-linked signal, not to establish a clinical-use algorithm. Apparent training performance was reported only as an audit trail and was not used as evidence of model reliability.

External Traceability Assessment

To assess transferability beyond TCGA, we evaluated the feature set in GSE31210 lung adenocarcinoma samples with complete follow-up (N=178). Cross-platform harmonization used feature alignment, dataset-specific normalization, batch adjustment where applicable, and within-cohort z-score scaling. Transferability was judged descriptively from discrimination, survival separation, and directionally compatible hazard ratios; these criteria were pragmatic and exploratory, not pre-registered clinical benchmarks.

Score-Behavior Metrics

Score behavior was assessed using AUC, accuracy, precision, recall, F1-score, C-index, hazard ratios, Kaplan–Meier curves, and exploratory decision-curve analysis where endpoint format allowed. Time-dependent ROC curves were generated at 1, 3, and 5 years. Median score splits were used only for visualization. Although XGBoost achieved the highest internal AUC, AdaBoost was retained as the reference score for downstream description because it produced stable survival separation and a compact descriptive score; no clinical superiority is claimed.

Clinical Comparator and Incremental-Value Boundaries

Where available, age, sex, and stage were included in multivariable survival models. However, harmonized PD-L1 immunohistochemistry, TMB, ctDNA dynamics, TCR repertoire, line of therapy, treatment regimen, and response-assessment metadata were not consistently available across the public cohorts. These variables were therefore not imputed or reconstructed for cross-cohort comparison. The study does not claim incremental predictive value over PD-L1, TMB, clinical-only models, stage-only models, or published immune signatures.

Statistical Analysis

Continuous variables were compared via Student's t-test or Mann-Whitney U test depending on normality (assessed by Shapiro-Wilk test). Categorical variables were compared using Chi-square test or Fisher's exact test. Survival analyses were performed using Cox proportional hazards regression, with both univariate and multivariate models adjusted for age, gender, and stage.

P < Correlation analyses used Pearson or Spearman correlation depending on distributional assumptions. Multiple testing correction used Benjamini-Hochberg FDR for differential-expression, MR, gene-marker correlation, and enrichment analyses; spatial autocorrelation used Bonferroni correction where specified. All statistical analyses were performed in R version 4.2.0 and Python 3.9.0. Two-sided P-values < 0.05 were considered statistically significant unless otherwise specified.

Hazard Ratio Calculation

Hazard ratios (HRs) and 95% confidence intervals (CIs) were calculated using univariate Cox proportional hazards regression implemented in the *lifelines* Python package (version 0.27.0). Risk scores from each machine learning model were obtained through 5-fold cross-validation, standardized using z-score transformation, and treated as continuous variables. HRs represent the hazard increase per 1 standard deviation increase in risk score. Wald 95% confidence intervals were computed, and P-values were obtained from the likelihood ratio test. For visualization purposes in Kaplan-Meier curves, patients were stratified into high-risk and low-risk groups based on the median risk score, and survival differences were assessed using the log-rank test.

Descriptive Score Construction and Survival Association

To evaluate whether the stability-prioritized feature set carried retrospective outcome signal, we constructed a descriptive outcome-associated score using AdaBoost with optimized hyperparameters ($n_estimators=100$, $learning_rate=0.1$, $random_state=42$). AdaBoost was used as a reference model for downstream description rather than as a clinically preferred algorithm. Patients were stratified into high-

and low-score groups using the median risk score for visualization only. Kaplan–Meier curves and log-rank tests were used to display survival differences, and continuous standardized scores were used in Cox models. Discrimination was summarized with C-index and time-dependent AUC.

Results

The results follow the declared evidence hierarchy: first establishing cohort accounting and feature stability, then asking whether the same macrophage-context signal is supported by MR-based prioritization, spatial organization, retrospective outcome traceability, and exploratory immunotherapy-response association.

Study Population and Baseline Characteristics

We initially considered 1,100 TCGA NSCLC tumor samples (595 LUAD, 505 LUSC). After quality control and availability filtering, 1,038 samples contributed to discovery and feature-screening analyses. The GSE31210 cohort included 226 tumor samples, of which 178 had complete follow-up for survival analysis. Key baseline differences included:

Gender disparity: LUSC had higher male representation (74.1% vs. 41.3% in LUAD)

Survival outcomes: LUSC patients showed higher mortality (43.6% vs. 32.1% in LUAD)

Detailed characteristics are provided in Table 2; cohort accounting and source-data traceability are documented in the public source-data manifest. PD-L1, TMB, ctDNA, TCR, and harmonized treatment-regimen annotations were not consistently available across these datasets; therefore, the manuscript does not present a formal incremental-value analysis over standard clinical or molecular comparators.

Table 2. Baseline characteristics of study population (TCGA NSCLC cohort and GSE31210 traceability cohort)

Characteristic	Total (N=1100)	LUAD (N=595)	LUSC (N=505)	GSE31210 (N=226 ^b)	P-value ^a
Age (years)	66.2 ± 9.4	65.1 ± 10.1	67.3 ± 8.6	59.6 ± 7.4	<0.001
Gender					0.001
Male	620 (56.4%)	246 (41.3%)	374 (74.1%)	105 (46.5%)	
Female	417 (37.9%)	286 (48.1%)	131 (25.9%)	121 (53.5%)	
Unknown	63 (5.7%)	63 (10.6%)	0 (0.0%)	0 (0.0%)	
AJCC Stage					<0.001
Stage I	535 (48.6%)	289 (48.6%)	246 (48.7%)	168 (74.3%)	
Stage II	288 (26.2%)	125 (21.0%)	163 (32.3%)	58 (25.7%)	
Stage III	169 (15.4%)	84 (14.1%)	85 (16.8%)	0 (0.0%)	
Stage IV	33 (3.0%)	26 (4.4%)	7 (1.4%)	0 (0.0%)	
Unknown	75 (6.8%)	71 (11.9%)	4 (0.8%)	0 (0.0%)	
Smoking History					N.A.
Light (<10 pack-years)	35 (3.2%)	22 (3.7%)	13 (2.6%)	N.A.	
Moderate (10–30 pack-years)	170 (15.5%)	105 (17.6%)	65 (12.9%)	N.A.	
Heavy (≥30 pack-years)	588 (53.5%)	238 (40.0%)	350 (69.3%)	N.A.	
Unknown	307 (27.9%)	230 (38.7%)	77 (15.2%)	N.A.	

Characteristic	Total (N=1100)	LUAD (N=595)	LUSC (N=505)	GSE31210 (N=226 ^b)	P-value ^a
Vital Status					<0.001
Alive	626 (56.9%)	341 (57.3%)	285 (56.4%)	191 (84.5%)	
Dead	411 (37.4%)	191 (32.1%)	220 (43.6%)	35 (15.5%)	
Unknown	63 (5.7%)	63 (10.6%)	0 (0.0%)	0 (0.0%)	

Note: Values are shown as mean $\pm$ SD or n (%). ^aP-values compare TCGA total versus GSE31210 using the Mann-Whitney U test for continuous variables and chi-square test for categorical variables. ^bThe GSE31210 cohort includes 226 primary lung adenocarcinoma samples after excluding normal tissue controls; survival analyses used 178 patients with complete follow-up data. N.A., not available.

Table 3. Retrospective score-behavior comparison across candidate algorithms

Model	Test AUC	Test Accuracy	Test F1-score	C-index	HR (95% CI)	P-value
XGBoost	0.781	0.725	0.715	0.591	1.56 (1.33–1.84)	<0.001
Naive Bayes	0.680	0.679	0.680	0.680	1.59 (1.33–1.91)	<0.001
Soft Voting	0.757	0.738	0.732	0.604	1.59 (1.35–1.88)	<0.001
Random Forest	0.701	0.657	0.486	0.657	1.59 (1.34–1.89)	<0.001
SVM	0.727	0.667	0.493	0.656	1.51 (1.28–1.79)	<0.001
Logistic Regression	0.741	0.705	0.652	0.650	1.45 (1.23–1.71)	<0.001
Gradient Boosting	0.742	0.698	0.685	0.650	1.56 (1.33–1.84)	<0.001
AdaBoost	0.743	0.657	0.526	0.607	1.43 (1.21–1.70)	<0.001
K-Nearest Neighbors	0.689	0.654	0.645	0.620	1.37 (1.15–1.64)	<0.001
Decision Tree	0.612	0.598	0.601	0.564	1.29 (1.10–1.51)	0.002
LASSO Cox	0.725	0.686	0.629	0.558	1.52 (1.39–1.66)	<0.001

Note: Models were evaluated using 5-fold nested cross-validation. Test AUC represents area under the ROC curve on the held-out fold. C-index measures risk ranking. HR and 95% CI were calculated using continuous Cox regression with standardized risk scores. Bold values identify the reference AdaBoost score used for downstream descriptive analyses; they do not indicate uniform superiority across all metrics or clinical readiness.

Robust Feature Selection via Nested Cross-Validation

Nested cross-validation identified 91 stable features selected in 100% of outer folds, with 212 features showing >60% selection frequency (Supplementary Figure S2, Supplementary Table S4). These features were used as a stability-prioritized set for downstream exploratory score construction. Among recurrent candidates, *ACP5*, *KCNAB2*, and *BOK* provided macrophage-regulatory, ion-channel-related, and apoptotic-context signals, respectively. Feature selection stability analysis confirmed separation between highly stable and less consistent features (Supplementary Figure S2A).

Mendelian Randomization Prioritizes a Candidate Gene Set in NSCLC

Mendelian randomization prioritized **198 genes as expression-linked candidates** associated with lung-cancer immune-context phenotypes (Figure 2, Supplementary Figures S9–S10, Supplementary Tables S7 and S11). The diagnostics suggested that some candidates had usable instruments, while others should be treated cautiously because of heterogeneity, pleiotropy, limited colocalization support, or bidirectional signals. These results are better viewed as a prioritization layer than as definitive causal proof.

MR-based Gene Prioritization

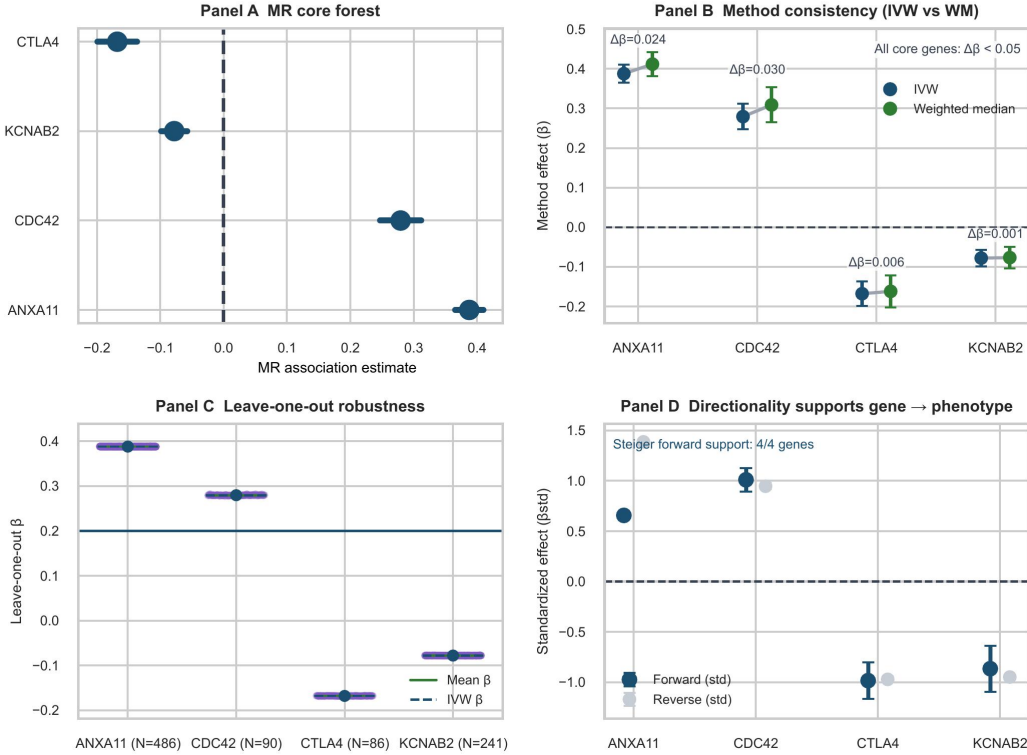


Figure 2. MR-based gene prioritization. Panel A, MR prioritizes C1Q-relevant candidate axes associated with immune phenotype. Panel B, Sensitivity analyses using IVW, MR-Egger, and weighted median estimates. Panel C, Leave-one-out analysis assesses single-SNP influence. Panel D, Bidirectional and directionality checks identify candidates requiring cautious interpretation.

KCNAB2, C1QA, BOK, ACP5, and HIST1H4F were among the better-ranked expression-linked candidates. Leave-one-out analyses suggested that the highlighted signals were not dominated by a single instrument. Functional enrichment of the prioritized set was strongest for immune-response, T-cell receptor, cytokine-receptor, and macrophage-context pathways. These outputs were used to rank candidates for follow-up and were not interpreted as quantitative evidence of mechanism.

Patient-Limited Spatial Mapping of C1Q/SPP1 Context

Spatial mapping supports non-random C1Q clustering in the available sections. Using both 10x Visium and CosMx SMI platforms, we observed that C1Q-family expression was spatially clustered rather than uniformly distributed within the available TME sections (Figures 3–4, Table 4, Supplementary Table S5). C1Q family members exhibited moderate spatial autocorrelation among immune genes:

$P < 0.001$ C1QA: Moran's I = 0.386 ($P < 0.001$)

$P < 0.001$ C1QC: Moran's I = 0.377 ($P < 0.001$)

$P < 0.001$ C1QB: Moran's I = 0.355 ($P < 0.001$)

Global analysis of 100 immune-related genes placed C1Q family members among the more spatially clustered immune genes, alongside UBXN10 (Moran's $I = 0.416$). SPP1 showed stronger spatial autocorrelation (Moran's $I = 0.712$, $P < 0.001$), consistent with a more spatially compartmentalized macrophage context.

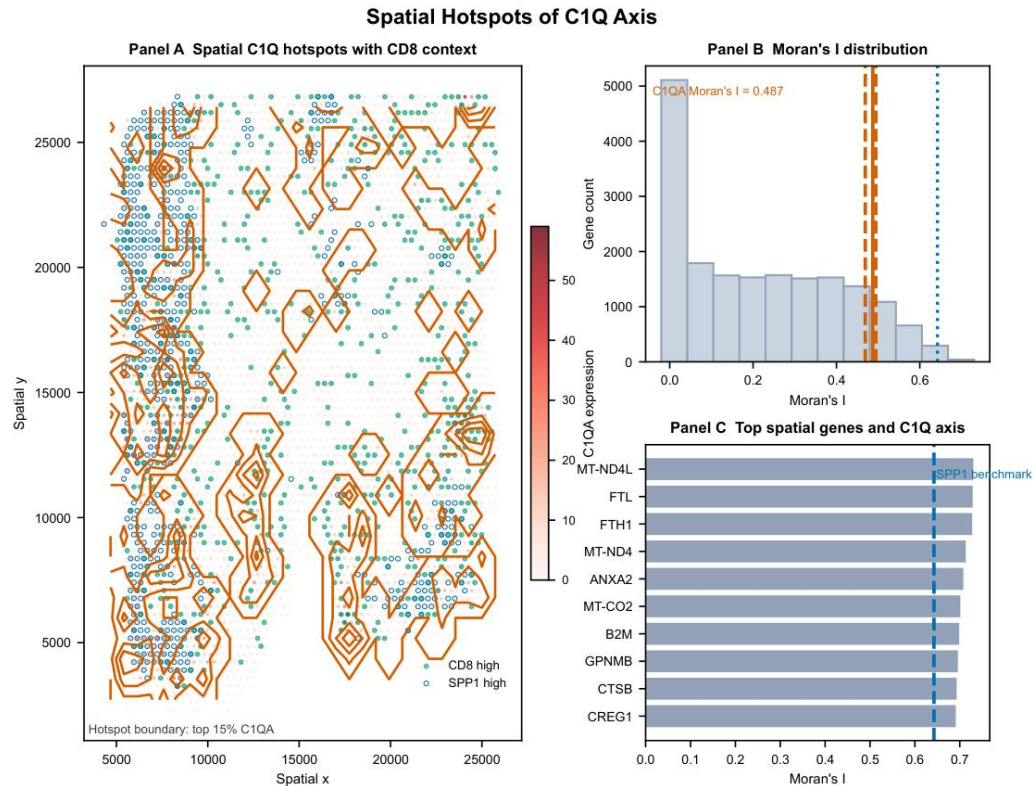


Figure 3. Spatial clustering of the C1Q/SPP1 axis. Panel A, Spatial clustering of C1Q expression across tumor sections. Panel B, C1Q spatial autocorrelation quantified by Moran's I . Panel C, Comparison with SPP1-enriched macrophage compartments shows distinct spatial organization. Panel D, CosMx single-cell spatial data provide context-level support for conservation of the pattern.

Neighborhood enrichment analysis was used to describe local co-occurrence between C1Q-associated macrophage contexts and CD8-marker-positive contexts. The pairwise neighborhood contrasts shown in Figure 4B were interpreted descriptively rather than as statistically significant enrichment. Separately, GigaTIME deconvolution linked C1Q regions to myeloid and T-cell marker enrichment, including CD14, CD163, CD68, and CD8-marker proxies (Figure 4, Supplementary Figure S6, Supplementary Table S6). These spot/cell-level summaries were treated as marker-context evidence rather than patient-level inference. CD138-positive plasma-cell context was analyzed separately and is not treated as a macrophage marker.

CosMx SMI context analysis across 8 sections from 5 patients showed broadly similar spatial patterns, although the cross-platform statistic was small in magnitude (bivariate Moran's $I = 0.009$). This supports limited reproducibility of the pattern rather than strong effect size.

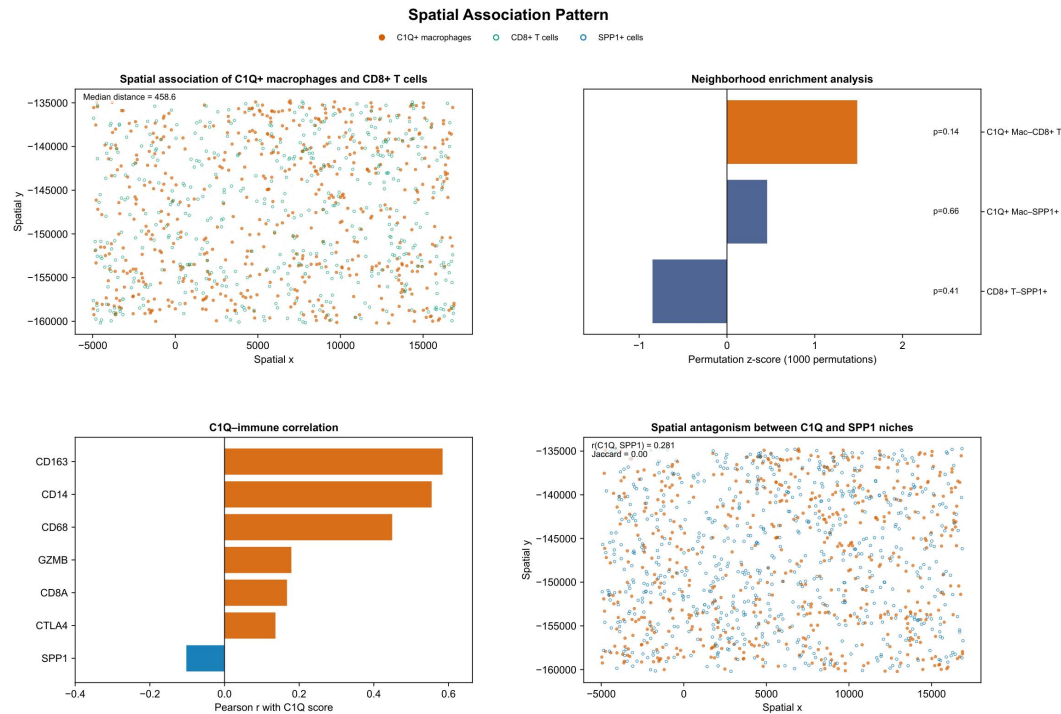


Figure 4. Spatial context of C1Q-associated macrophage states. Panel A, Local proximity of C1Q-marker-positive macrophage contexts and CD8-marker-positive contexts. Panel B, Neighborhood enrichment analysis summarizes attraction/avoidance structure. Panel C, Spatial correlation between C1Q and immune markers supports context-level association. Panel D, Spatial contrast between C1Q- and SPP1-enriched niches delineates immune-active versus immune-excluded states.

Counterfactual Scoring Shows Dependency of the C1Q Module

To test whether the C1Q score was internally dependent on C1QA, we performed a counterfactual scoring analysis and re-estimated spatial and clinical metrics (Figure 5, Supplementary Figures S11–S12). The computational attenuation reduced the C1Q program score (mean C1Q score: 5.179 $\rightarrow$ 2.628) and decreased Moran's I (0.660 $\rightarrow$ 0.401; $\Delta = -0.259$). This indicates score dependency and module sensitivity, not experimental proof of biological perturbation.

Additional sensitivity analyses showed a coherent descriptive pattern across several views. Dose-response modeling showed monotonic shifts across C1Q quartiles, bidirectional MR identified forward and reverse signals for some candidates, spatial-scale analysis suggested the strongest local association at the shortest tested radius, and specificity testing ranked the C1Q axis among the better-supported immune modules relative to non-C1Q controls (Supplementary Figure S12). These analyses support internal consistency of the score and do not establish biological perturbation.

The main visualization is summarized in Figure 5, with additional network-summary sensitivity outputs provided in Supplementary Figures S11–S12. These analyses are sensitivity checks rather than experimental perturbations.

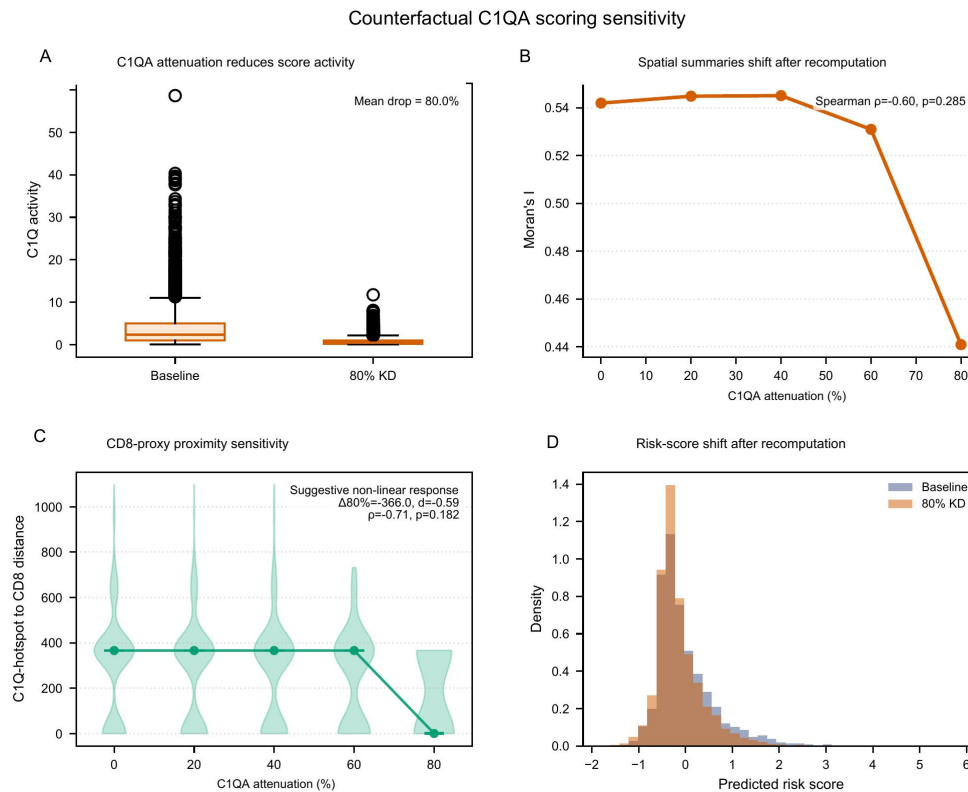


Figure 5. Counterfactual scoring sensitivity of the C1Q axis. Panel A, Computational C1Q-axis attenuation reduces composite axis activity. Panel B, Spatial hotspot structure shifts after score recomputation. Panel C, CD8-proxy proximity summaries show threshold-like sensitivity to C1Q-axis attenuation. Panel D, Risk-score distributions shift after counterfactual score recomputation. These analyses test internal score dependency and should not be interpreted as experimental gene attenuation or causal perturbation.

Retrospective Score Behavior and Outcome Traceability

The C1Q/SPP1 feature set showed retrospective survival-associated score behavior. Evaluation across candidate algorithms showed that different methods optimized different metrics (Table 3, Figure 6, Supplementary Figures S2 and S8, Supplementary Table S1). AdaBoost was retained as the reference score for downstream description, but it was not uniformly superior across all metrics.

Key retrospective metrics. AdaBoost showed moderate ranking performance (C-index = 0.607) and an association between standardized risk score and survival (HR = 1.430, 95% CI: 1.21–1.70, $P < 0.001$). XGBoost achieved the highest internal AUC (0.781), whereas other algorithms showed higher C-index values. This metric discordance argues against presenting any single model as clinically superior.

Score-Group Separation: AdaBoost separated a TCGA complete-case subset into high-score and low-score groups with different survival outcomes (log-rank $P = 0.005$, HR = 1.620, 95% CI: 1.15–2.27). To avoid ambiguity, higher model risk score denotes the adverse-risk direction; C1Q-enriched immune-active context is interpreted separately from the aggregate risk-score direction. Time-dependent ROC showed modest discrimination: 1-year AUC of 0.609, 3-year AUC of 0.611, and 5-year AUC of 0.590 (Figure 6A–B). Risk-score ordering, survival status, and feature-expression patterns are shown as

descriptive internal score-behavior summaries (Figure 6C–E), and decision-curve analysis is reported only as exploratory net-benefit visualization (Figure 6F).

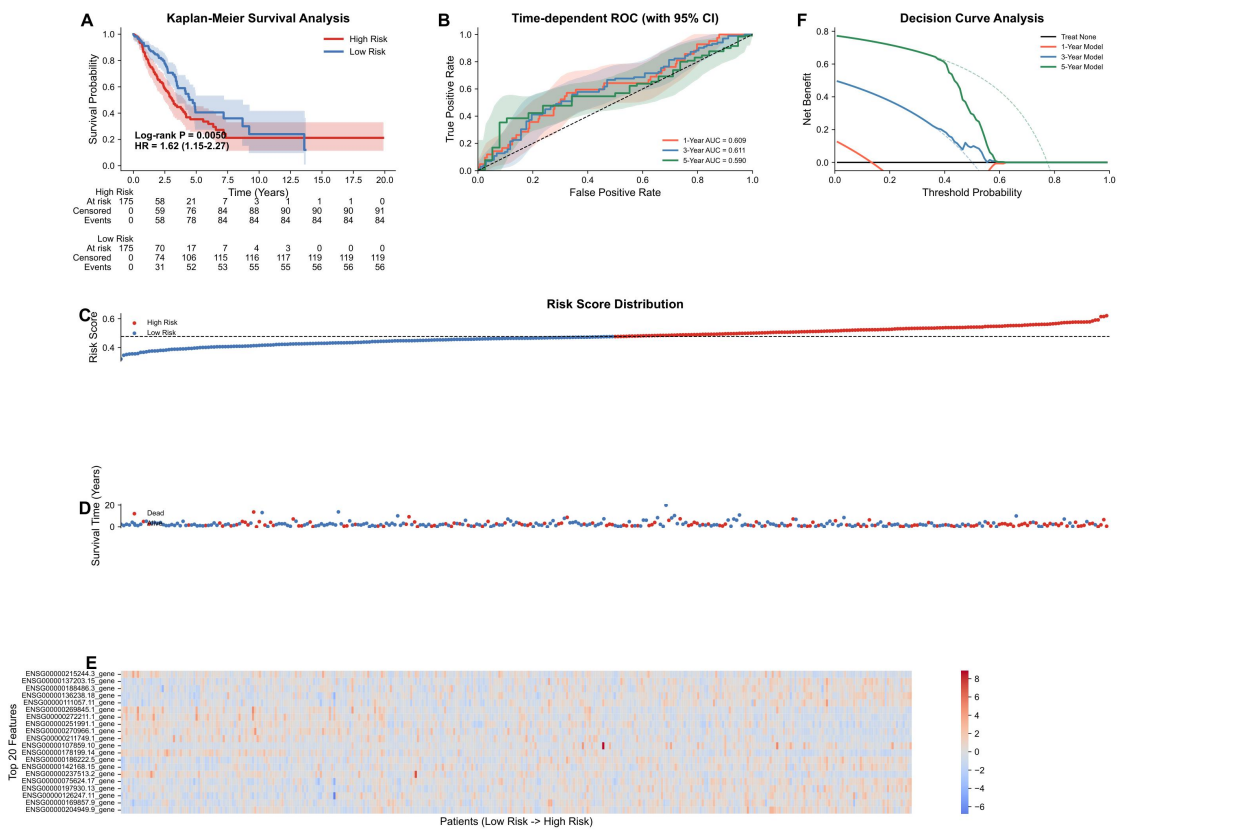


Figure 6. Retrospective score behavior of the C1Q-centered feature set. Panel A, Kaplan–Meier analysis shows survival separation between high- and low-score groups in a TCGA complete-case subset. Panel B, Time-dependent ROC curves quantify discrimination at 1, 3, and 5 years. Panels C–E, Risk-score ordering, survival-status distribution, and feature-expression heatmap summarize internal score behavior across patients. Panel F, Decision curve analysis provides exploratory net-benefit visualization and is not interpreted as clinical utility evidence.

External Traceability: In GSE31210, the feature set showed attenuated but directionally compatible score behavior (AUC = 0.650, C-index = 0.629; Figure 7, Supplementary Figure S7, Supplementary Table S13). Survival separation was sensitive to analysis specification, with source tables reporting both significant and borderline estimates depending on completeness and covariate handling. Therefore, GSE31210 is treated as external traceability rather than definitive evidence. Decision curve analysis is reported only as exploratory net-benefit visualization.

External traceability of C1Q/SPP1 score behavior

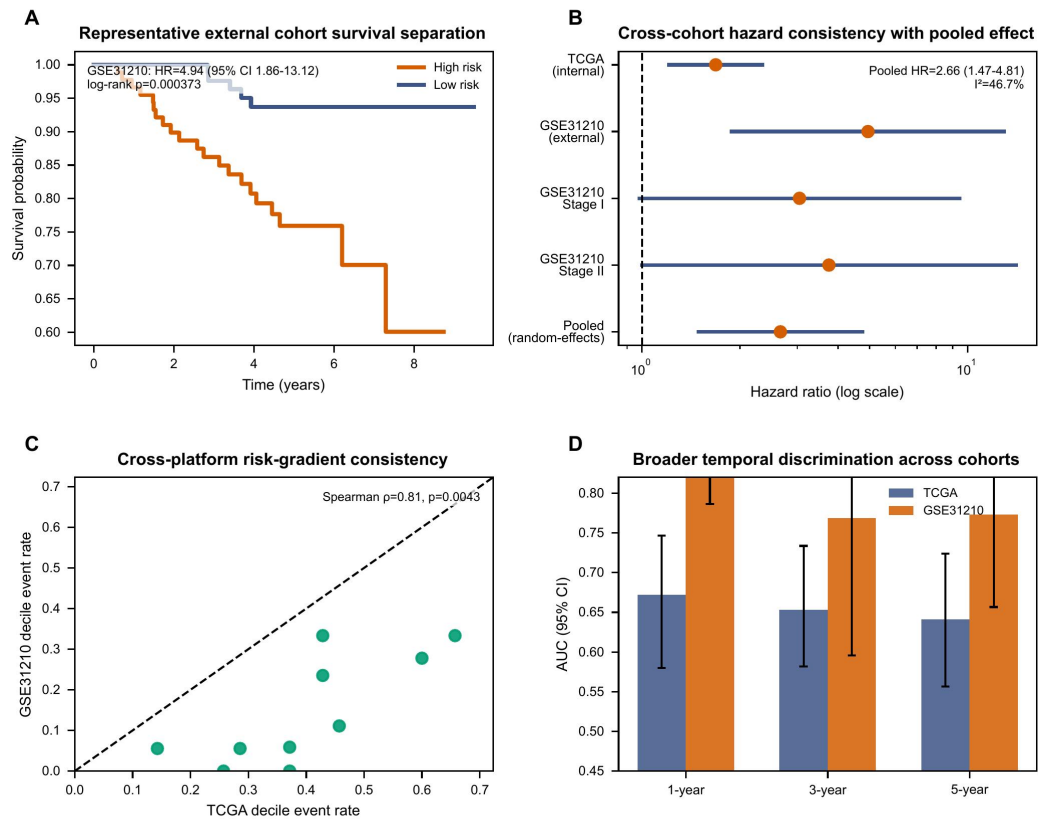


Figure 7. External traceability and transferability. Panel A, Kaplan–Meier display in the GSE31210 traceability cohort summarizes score-group separation. Panel B, Cross-cohort discrimination metrics summarize transferability. Panel C, Cross-platform risk-score consistency is shown as technical traceability. Panel D, Pan-cancer C1Q-axis patterns are interpreted as exploratory biological context. These panels support external traceability rather than definitive evidence.

Exploratory Immunotherapy-Response Association

We next examined whether the C1Q/SPP1-associated score varied with response annotations in small immunotherapy-treated datasets. These analyses are exploratory because cohort sizes were limited and GSE91061 is not NSCLC-specific. The cohorts were not used to tune the score or select a response threshold. We therefore emphasize direction and consistency rather than presenting these cohorts as confirmation of ICB response utility.

In GSE91061, a melanoma cohort, lower-score patients showed directionally higher response rates and responders showed a decrease in score during therapy; because this cohort is not NSCLC-specific, it is treated only as cross-tumor exploratory context. In the two small NSCLC ICB cohorts, GSE126044 (N=16) and GSE135222 (N=27), the score showed response-associated separation in the expected direction, but precision was limited and no claim is made regarding clinical response prediction or superiority over PD-L1, TMB, or clinical baselines.

Subtype summaries suggested that lower-score or C1Q-enriched contexts were more often aligned with inflamed immune programs, whereas higher-score contexts were more often aligned with excluded or desert-like features, including SPP1-associated macrophage programs. These subtype assignments are descriptive and require independent evaluation.

Multivariable Survival Association

To assess whether the score retained survival association after adjustment for available clinical variables, we performed Cox regression analyses (Table 5, Supplementary Table S13).

In TCGA complete-case models, the standardized risk score retained an association with overall survival after adjustment for age, gender, and stage (multivariable HR = 1.430, 95% CI: 1.21–1.70; Table 5).

These analyses remain retrospective and should be interpreted as outcome traceability rather than clinical evidence. External covariate summaries and GSE31210 traceability analyses are provided in Supplementary Tables S9 and S13.

Integration of MR-Prioritized Genes and Spatial Correlations

The 91-feature stability set, the 198-gene MR-prioritized set, and the spatially clustered genes were not treated as interchangeable evidence layers. Instead, convergence across these layers was used to rank follow-up candidates (Table 4). KCNAB2 was retained as a recurrent candidate because it appeared in the stable feature set, was prioritized by MR, and showed measurable but modest spatial-context support. SPP1 and the C1Q-family genes provided the clearest macrophage-context contrast, whereas CDC42, PLOD1, LINC00339, and UBXN10 were treated as secondary network or spatial-context candidates.

Table 4. Cross-layer evidence map for prioritized candidate genes

Gene	Annotation	Selection Stability	Moran's I	Network Degree	Correlation (r)
ACP5	Macrophage Reg.	100% (5/5)	N.D.	-	-
KCNAB2	Ion Channel	100% (5/5)	0.094***	0.078	0.219***
SPP1	TME Modulation	-	0.712***	0.066	-0.258***
LINC00339	Long ncRNA	-	N.D.	0.064	0.586***
CDC42	Signal Transduction	-	N.D.	0.060	0.424***
PLOD1	Collagen Modification	-	N.D.	0.035	0.260***
C1QA	Complement	-	0.386***	-	-
C1QB	Complement	-	0.353***	-	-
C1QC	Complement	-	0.377***	-	-
UBXN10	Ubiquitination	-	0.416***	-	-
BOK	Apoptosis	100% (5/5)	N.D.	-	-

Note: N.D., not determined. Selection stability indicates membership in the 91-feature stability-prioritized set. Network degree centrality derives from the MR-prioritized network. Correlation represents association with immune-context estimates. These entries identify candidates for follow-up and should not be read as causal, spatial, or clinical confirmation.

Table 5. Univariate and multivariable Cox regression analysis in the TCGA discovery cohort (N = 1,038)

	Univariate analysis		Multivariable analysis	
Covariate	HR (95% CI)	P-value	HR (95% CI)	P-value
Risk Score ^a	2.45 (1.85–3.24)	<0.001	1.43 (1.21–1.70)	<0.001
Age	1.02 (1.00–1.03)	0.064	1.01 (0.99–1.03)	0.217
Gender (Male vs Female)	0.98 (0.73–1.32)	0.898	0.90 (0.67–1.21)	0.482
Stage (IV vs I)	1.33 (1.09–1.61)	0.005	1.35 (1.11–1.65)	0.003

Note: Univariate and multivariable Cox regression analysis in the TCGA cohort included risk score, age, gender, and AJCC stage. Risk score was standardized before analysis. The HR represents the change in risk per standard-deviation increase in risk score. These retrospective associations require external and prospective testing before clinical interpretation. For GSE31210 traceability results, see Supplementary Table S13.

Public Protein-Expression Context

We cross-referenced public Human Protein Atlas immunohistochemistry images for protein-level context. KCNAB2 showed cytoplasmic staining in available lung tissue images, while SPP1 was predominantly stromal in the inspected examples (Figure 8). These public images support plausibility of protein-level expression patterns but should be read only as public protein-expression context, not as author-generated protein-level evidence. Source pages and license information are documented in the public source-data manifest.

Discussion

Principal Findings

This study supports a conservative conclusion: C1Q/SPP1 is a cross-dataset macrophage-context traceability feature in public NSCLC resources. Across bulk transcriptomics, MR-based prioritization, spatial datasets, and retrospective outcome cohorts, the C1Q/SPP1 axis repeatedly marked immune-active versus macrophage-compartmentalized states. The evidence is internally coherent, but it does not establish a clinical assay, a treatment-selection rule, or a definitive mechanism.

The principal findings were: (1) nested cross-validation identified a stable 91-feature set enriched for immune and macrophage-context signals; (2) MR prioritized several candidates, including *C1QA* and *KCNAB2*, but with the usual instrumental-variable caveats; (3) C1Q-family genes showed moderate spatial clustering in patient-limited spatial data, while *SPP1* showed stronger spatial compartmentalization; (4) counterfactual scoring showed that the C1Q module is internally sensitive to *C1QA* attenuation, without proving biological perturbation; and (5) retrospective score behavior was statistically detectable but modest, with external and immunotherapy-response results best interpreted as traceability rather than confirmation.

Interpretation of the Evidence Chain

The strength of the study is not a single high-performing classifier. Rather, it is the convergence of several public-data views on the same macrophage-context axis. Bulk expression suggested a survival-associated immune program; MR prioritized genes that intersected with this program; spatial data placed C1Q and SPP1 in non-random local contexts; and small immunotherapy-treated cohorts showed directional response associations. This layered consistency makes the C1Q/SPP1 axis biologically plausible and worth further study, but it should be understood as traceability rather than validation.

At the same time, each layer has specific inference limits. MR prioritizes expression-linked candidates under instrumental-variable assumptions but does not establish mechanism. Spatial autocorrelation describes local organization but does not prove cell-cell communication. Computational C1QA attenuation tests score dependency but is not a wet-lab perturbation. Small ICB cohorts provide response-association context but not clinical evidence. For that reason, the manuscript should be read as a traceable, hypothesis-generating cancer-informatics study rather than as a completed biomarker-development paper.

Biological Plausibility of the C1Q/SPP1 Spatial Context

The biological plausibility of the C1Q/SPP1 axis rests on macrophage biology and spatial organization. C1Q-associated regions tracked immune-marker context, suggesting that complement-associated macrophage states may mark immune-active local environments; this is a marker-context interpretation, not proof that C1Q drives therapeutic response. The stronger spatial autocorrelation of *SPP1* (Moran's $I = 0.712$) delineated a distinct macrophage-associated compartment compatible with stromal-remodeling or immune-exclusion context, but it does not prove that *SPP1* creates a physical barrier in NSCLC.

Additional candidates such as *BOK* and *HIST1H4F* were retained only as follow-up hypotheses because they intersected with feature stability, MR prioritization, or functional context. Together, these patterns

argue that bulk expression signatures should be interpreted with spatial and cellular context, but they do not yet justify a diagnostic or therapeutic strategy.

KCNAB2 as a Candidate Context Feature

KCNAB2, a potassium-channel subunit, emerged as a recurrent candidate in feature selection and MR-prioritized analyses. Its co-occurrence with C1Q-associated immune contexts raises the hypothesis that ion-channel-related programs may track macrophage or immune-effector state, but the present data cannot determine cell type of action or functional direction. We therefore describe *KCNAB2* as a candidate context feature, not a metabolic checkpoint, therapeutic target, or spatial organizer. Its status should be tested using cell-type-resolved protein assays, perturbation experiments, and prospective outcome cohorts.

Machine Learning Model Behavior and Its Limits

Score behavior depended on the metric used. XGBoost achieved the highest internal AUC, whereas other models had comparable or higher ranking metrics; therefore, the algorithm comparison should not be read as evidence of clinical superiority. AdaBoost was retained only as a reference model for describing one internally traceable score.

The model captured statistically detectable but modest prognostic signal and should not be used for individual patient decision-making. Decision-curve analysis is included only as exploratory visualization. External GSE31210 results were directionally supportive but not definitive, and the small ICB cohorts cannot establish ICI response-associated score utility or superiority over PD-L1, TMB, clinical-only models, stage-only models, or published immune signatures.

Future Translation Requirements

Future translation would require a locked feature set, preprocessing pipeline, model, and cutoff before testing; adequately powered NSCLC-only immunotherapy cohorts; patient- or section-level spatial summaries that avoid cell-count inflation; comparison with stage-only, clinical-only, PD-L1, TMB, and published immune-signature baselines; multiplex FFPE-compatible spatial protein confirmation; and experimental testing of whether C1Q/SPP1-associated macrophage states influence T-cell localization or activity. Only after these steps would a simplified pathological or molecular panel be justified.

Limitations and Future Directions

Several limitations require emphasis. The primary survival analyses are retrospective and rely heavily on overall survival, which is affected by treatment heterogeneity, comorbidity, stage, and incomplete clinical annotation. The spatial datasets include few independent patients, and the large CosMx cell count should not be mistaken for a large number of independent biological replicates. The Visium dataset provides spatial context but not single-cell precision, and the available spatial data do not support a locked patient-level spatial classifier. The GSE91061 cohort is melanoma rather than NSCLC, and the NSCLC ICB cohorts are small. Integration with PD-L1, TMB, ctDNA dynamics, TCR repertoire, line of therapy, treatment regimen, and response-assessment metadata will be needed before incremental value can be assessed. Additional limitations include cross-platform technical variance, possible residual confounding, and the absence of direct wet-lab functional assays. The model has not been tested in neoadjuvant

immunotherapy, prospective randomized settings, or an independently locked NSCLC ICB response cohort.

Overall, the study provides a public-data framework for tracing macrophage-context signals across molecular and spatial layers. Its most immediate value is hypothesis prioritization for future spatial and functional testing.

Conclusions

This study identifies C1Q/SPP1 as a cross-dataset macrophage-context traceability feature in public NSCLC resources. The principal contribution is not a deployable classifier, but a transparent evidence chain linking feature stability, MR-based prioritization, patient-limited spatial organization, and cautious outcome traceability. C1Q-family genes showed moderate spatial clustering, SPP1 showed stronger spatial compartmentalization, and retrospective score behavior suggested modest outcome-linked signal. These findings prioritize a macrophage-context axis for prospective, NSCLC-specific, spatially resolved evaluation and experimental testing. They should not be interpreted as a validated biomarker, treatment-selection rule, causal mechanism, or therapeutic-target claim.

Supplementary Information

Additional file 1: Supplementary Material. Contains Supplementary Figures S1–S12 and Supplementary Tables S1–S13, including feature-selection stability, MR diagnostics, spatial summaries, model-performance tables, external-cohort covariate summaries, and reporting-boundary tables.

Statements and Declarations

Ethics Approval and Consent to Participate

This study utilized publicly available datasets from The Cancer Genome Atlas (TCGA), Gene Expression Omnibus (GEO: GSE31210, GSE91061, GSE126044, GSE135222), eQTLGen Consortium, GTEx Consortium, and Human Protein Atlas. All original studies from which these data were derived obtained appropriate informed consent from participants and received approval from their respective institutional review boards. As this study represents a secondary analysis of de-identified public data, additional ethics approval was not required in accordance with institutional guidelines.

Consent for Publication

Not applicable. This manuscript does not contain identifiable individual-person data. Public Human Protein Atlas images are used only as non-identifiable protein-expression context and are cited with source and license information.

Availability of Data and Materials

Publicly available datasets were analyzed in this study. These data can be found here:

The Cancer Genome Atlas (TCGA) NSCLC cohort data: NCI Genomic Data Commons (GDC) Portal (<https://portal.gdc.cancer.gov>).

Gene Expression Omnibus datasets: NCBI GEO.⁵² <https://www.ncbi.nlm.nih.gov/geo/>.

GSE31210: Lung adenocarcinoma traceability cohort (N=226 accession-level samples; N=178 analyzed with complete follow-up)

GSE91061: Melanoma immunotherapy cohort (N=109 samples)

GSE126044: NSCLC anti-PD-1 therapy cohort (N=16 patients)

GSE135222: NSCLC PD-1/PD-L1 inhibitor cohort (N=27 patients)

eQTL summary statistics: eQTL Catalogue (<https://www.ebi.ac.uk/eql/>) and eQTLGen Consortium (<https://www.eqtlgen.org/>).

GTEx v8 data: GTEx Portal (<https://gtexportal.org/>).

Immunohistochemistry images: Human Protein Atlas.⁵³ <https://www.proteinatlas.org/>; figure-specific source and license notes are provided in the source-data manifest.

Code Availability

The source-data manifest, processed analysis tables, manuscript files, figures, and reproducibility notes are available in a public GitHub repository: <https://github.com/morningLxj/spatial-immunotherapy-prediction>. The repository includes a machine-readable source-data archive and claim-to-source mapping for reviewer inspection. It is intended to support traceability and source-data verification, not to provide a clinical implementation package or a complete raw-data-to-figure re-analysis pipeline.

Repository integrity checks and reproducibility notes are available at <https://github.com/morningLxj/spatial-immunotherapy-prediction>. The included script verifies the expected source-data archive, lists its contents, and writes a source-data inventory. Public-data acquisition and complete re-analysis from raw source repositories should be treated as a separate workflow and, if required by the journal or reviewers, archived as a versioned release before final publication.

Reporting Guidelines

The manuscript was revised using STROBE principles for observational retrospective analyses and TRIPOD-AI considerations where applicable. Because the work is a public-data evidence synthesis rather than a locked clinical model-development study, the model results are presented as exploratory score behavior.

Declaration of Conflicting Interests

The authors declare that they have no competing interests, financial or otherwise, that could influence the work reported in this paper.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The authors declare that no funds, grants, or other financial support were received during the preparation of this manuscript.

Authors' Contributions

Xiongjie Li conceived and supervised the study, contributed to study design, and revised the manuscript. Fengyue Zhang, Chaojia Luo, and Xuan Xu contributed to data interpretation, manuscript drafting or revision, and critical review. All authors read and approved the final manuscript.

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Acknowledgements

The authors thank the investigators and participants of the public datasets and resources used in this study, including TCGA, GEO, eQTLGen, GTEx, and the Human Protein Atlas.

References

1. Sung H, Ferlay J, Siegel RL, et al. Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. *CA Cancer J Clin*. 2021;71(3):209-249. doi:10.3322/caac.21660
2. Chen Z, Fillmore CM, Hammerman PS, Kim CF, Wong KK. Non-small-cell lung cancers: a heterogeneous set of diseases. *Nat Rev Cancer*. 2014;14(8):535-546. doi:10.1038/nrc3775
3. Reck M, Rodríguez-Abreu D, Robinson AG, et al. Pembrolizumab versus Chemotherapy for PD-L1–Positive Non–Small-Cell Lung Cancer. *N Engl J Med*. 2016;375(19):1823-1833. doi:10.1056/nejmoa1606774
4. Herbst RS, Giaccone G, de Marinis F, et al. Atezolizumab for First-Line Treatment of PD-L1–Selected Patients with NSCLC. *N Engl J Med*. 2020;383(14):1328-1339. doi:10.1056/nejmoa1917346
5. Gandhi L, Rodríguez-Abreu D, Gadgeel S, et al. Pembrolizumab plus Chemotherapy in Metastatic Non–Small-Cell Lung Cancer. *N Engl J Med*. 2018;378(22):2078-2092. doi:10.1056/nejmoa1801005
6. Passaro A, Attili I, de Marinis F. Neoadjuvant Chemotherapy Plus Immunotherapy in Early-Stage Resectable Non–Small-Cell Lung Cancer. *J Clin Oncol*. 2022;40(25):2871-2877. doi:10.1200/jco.22.00873
7. Hellmann MD, Ciuleanu TE, Pluzanski A, et al. Nivolumab plus Ipilimumab in Lung Cancer with a High Tumor Mutational Burden. *N Engl J Med*. 2018;378(22):2093-2104. doi:10.1056/nejmoa1801946
8. Marabelle A, Fakih M, Lopez J, et al. Association of tumour mutational burden with outcomes in patients with advanced solid tumours treated with pembrolizumab: prospective biomarker analysis of the multicohort, open-label, phase 2 KEYNOTE-158 study. *Lancet Oncol*. 2020;21(10):1353-1365. doi:10.1016/s1470-2045(20)30445-9
9. Topalian SL, Taube JM, Anders RA, Pardoll DM. Mechanism-driven biomarkers to guide immune checkpoint blockade in cancer therapy. *Nat Rev Cancer*. 2016;16(5):275-287. doi:10.1038/nrc.2016.36
10. Schoenfeld AJ, Hellmann MD. Acquired Resistance to Immune Checkpoint Inhibitors. *Cancer Cell*. 2020;37(4):443-455. doi:10.1016/j.ccell.2020.03.017
11. Vodnala SK, Eil R, Kishton RJ, et al. T cell stemness and dysfunction in tumors are triggered by a common mechanism. *Science*. 2019;363(6434):0135. doi:10.1126/science.aau0135
12. Vigneron N. Human Tumor Antigens and Cancer Immunotherapy. *BioMed Research International*. 2015;2015:1-17. doi:10.1155/2015/948501
13. Jerby-Arnon L, Shah P, Cuoco MS, et al. A Cancer Cell Program Promotes T Cell Exclusion and Resistance to Checkpoint Blockade. *Cell*. 2018;175(4):984-997.e24. doi:10.1016/j.cell.2018.09.006
14. Ståhl PL, Salmén F, Vickovic S, et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. *Science*. 2016;353(6294):78-82. doi:10.1126/science.aaf2403

15. Marx V. Method of the Year: spatially resolved transcriptomics. *Nat Methods*. 2021;18(1):9-14. doi:10.1038/s41592-020-01033-y
16. Zilionis R, Engblom C, Pfirschke C, et al. Single-Cell Transcriptomics of Human and Mouse Lung Cancers Reveals Conserved Myeloid Populations across Individuals and Species. *Immunity*. 2019;50(5):1317-1334.e10. doi:10.1016/j.immuni.2019.03.009
17. Laumont CM, Nelson BH. B cells in the tumor microenvironment: Multi-faceted organizers, regulators, and effectors of anti-tumor immunity. *Cancer Cell*. 2023;41(3):466-489. doi:10.1016/j.ccell.2023.02.017
18. Janesick A, Shelansky R, Gottscho AD, et al. High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and in situ analysis. *Nat Commun*. 2023;14(1):8353. doi:10.1038/s41467-023-43458-x
19. He K, Zhang X, Ren S, Sun J. Deep Residual Learning for Image Recognition. 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR). 2016:770-778. doi:10.1109/cvpr.2016.90
20. Campanella G, Hanna MG, Geneslaw L, et al. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nat Med*. 2019;25(8):1301-1309. doi:10.1038/s41591-019-0508-1
21. Valanarasu MJ, Xu H, Usuyama N, et al. Multimodal AI generates virtual population for tumor microenvironment modeling. *Cell*. 2026;189(2):386-400.e19. doi:10.1016/j.cell.2025.11.016
22. Wu Y, Yang S, Ma J, et al. Spatiotemporal Immune Landscape of Colorectal Cancer Liver Metastasis at Single-Cell Level. *Cancer Discovery*. 2022;12(1):134-153. doi:10.1158/2159-8290.cd-21-0316
23. Chen RJ, Lu MY, Williamson DFK, et al. Pan-cancer integrative histology-genomic analysis via multimodal deep learning. *Cancer Cell*. 2022;40(8):865-878.e6. doi:10.1016/j.ccell.2022.07.004
24. Thorsson V, Gibbs DL, Brown SD, et al. The Immune Landscape of Cancer. *Immunity*. 2018;48(4):812-830.e14. doi:10.1016/j.immuni.2018.03.023
25. Subramanian I, Verma S, Kumar S, Jere A, Anamika K. Multi-omics Data Integration, Interpretation, and Its Application. *Bioinform Biol Insights*. 2020;14:117793221989905. doi:10.1177/1177932219899051
26. Chaudhary K, Poirion OB, Lu L, Garmire LX. Deep Learning-Based Multi-Omics Integration Robustly Predicts Survival in Liver Cancer. *Clinical Cancer Res*. 2018;24(6):1248-1259. doi:10.1158/1078-0432.ccr-17-0853
27. Sammut SJ, Crispin-Ortuzar M, Chin SF, et al. Multi-omic machine learning predictor of breast cancer therapy response. *Nature*. 2022;601(7894):623-629. doi:10.1038/s41586-021-04278-5
28. Hoadley KA, Yau C, Hinoue T, et al. Cell-of-Origin Patterns Dominate the Molecular Classification of 10,000 Tumors from 33 Types of Cancer. *Cell*. 2018;173(2):291-304.e6. doi:10.1016/j.cell.2018.03.022
29. The Cancer Genome Atlas Research Network. Comprehensive molecular profiling of lung adenocarcinoma. *Nature*. 2014;511(7511):543-550. doi:10.1038/nature13385
30. The Cancer Genome Atlas Research Network. Comprehensive genomic characterization of squamous cell lung cancers. *Nature*. 2012;489(7417):519-525. doi:10.1038/nature11404
31. Thorsson V, Gibbs DL, Brown SD, et al. The Immune Landscape of Cancer. *Immunity*. 2018;48(4):812-830.e14. doi:10.1016/j.immuni.2018.03.023
32. Zhang L, Li Z, Skrzypczynska KM, et al. Single-Cell Analyses Inform Mechanisms of Myeloid-Targeted Therapies in Colon Cancer. *Cell*. 2020;181(2):442-459.e29. doi:10.1016/j.cell.2020.03.048
33. Davey Smith G, Ebrahim S. 'Mendelian randomization': can genetic epidemiology contribute to understanding environmental determinants of disease?*. *International Journal of Epidemiology*. 2003;32(1):1-22. doi:10.1093/ije/dyg070
34. Hagberg AA, Schult DA, Swart PJ. Exploring Network Structure, Dynamics, and Function using NetworkX. *Proceedings of the Python in Science Conference*. 2008:11-15. doi:10.25080/twv9851
35. Vösa U, Claringbould A, Westra HJ, et al. Large-scale cis- and trans-eQTL analyses identify thousands of genetic loci and polygenic scores that regulate blood gene expression. *Nat Genet*. 2021;53(9):1300-1310. doi:10.1038/s41588-021-00913-z
36. Zhu Z, Zhang F, Hu H, et al. Integration of summary data from GWAS and eQTL studies predicts complex trait gene targets. *Nat Genet*. 2016;48(5):481-487. doi:10.1038/ng.3538
37. McKinney W. Data Structures for Statistical Computing in Python. *Proceedings of the Python in Science Conference*. 2010:56-61. doi:10.25080/majora-92bf1922-00a
38. Harris CR, Millman KJ, van der Walt SJ, et al. Array programming with NumPy. *Nature*. 2020;585(7825):357-362. doi:10.1038/s41586-020-2649-2
39. Virtanen P, Gommers R, Oliphant TE, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat Methods*. 2020;17(3):261-272. doi:10.1038/s41592-019-0686-2
40. Hunter JD. Matplotlib: A 2D Graphics Environment. *Comput. Sci. Eng*. 2007;9(3):90-95. doi:10.1109/mcse.2007.55
41. Waskom M. seaborn: statistical data visualization. *JOSS*. 2021;6(60):3021. doi:10.21105/joss.03021

42. Pedregosa F, Varoquaux G, Gramfort A, et al. Scikit-learn: Machine learning in python. *J Mach Learn Res.* 2011;12:2825-2830.
43. Tang Z, Kang B, Li C, Chen T, Zhang Z. GEPIA2: an enhanced web server for large-scale expression profiling and interactive analysis. *Nucleic Acids Res.* 2019;47(W1):W556-W560. doi:10.1093/nar/gkz430
44. Hemani G, Zheng J, Elsworth B, et al. The MR-Base platform supports systematic causal inference across the human phenome. *eLife.* 2018;7:34408. doi:10.7554/elife.34408
45. Pan L, Luo J, Nie C, et al. A Deep Learning Framework Integrating Tumor Microenvironmental Features Accurately Predicts Multiple Driver Gene Mutations in Lung Cancer Pathology Images. *Cancer Res.* 2026;86(5):1319-1337. doi:10.1158/0008-5472.can-25-0582
46. Xu H, Wang S, Fang M, et al. SPACEL: deep learning-based characterization of spatial transcriptome architectures. *Nat Commun.* 2023;14(1):7603. doi:10.1038/s41467-023-43220-3
47. Ravi VM, Will P, Kueckelhaus J, et al. Spatially resolved multi-omics deciphers bidirectional tumor-host interdependence in glioblastoma. *Cancer Cell.* 2022;40(6):639-655.e13. doi:10.1016/j.ccell.2022.05.009
48. Li Q, Zhang X, Ke R. Spatial Transcriptomics for Tumor Heterogeneity Analysis. *Front Genet.* 2022;13. doi:10.3389/fgene.2022.906158
49. Petitprez F, de Reyniès A, Keung EZ, et al. B cells are associated with survival and immunotherapy response in sarcoma. *Nature.* 2020;577(7791):556-560. doi:10.1038/s41586-019-1906-8
50. Helmink BA, Reddy SM, Gao J, et al. B cells and tertiary lymphoid structures promote immunotherapy response. *Nature.* 2020;577(7791):549-555. doi:10.1038/s41586-019-1922-8
51. Rakaee M, Tafavvoghi M, Ricciuti B, et al. Deep Learning Model for Predicting Immunotherapy Response in Advanced Non-Small Cell Lung Cancer. *JAMA Oncol.* 2025;11(2):109. doi:10.1001/jamaoncol.2024.5356
52. Barrett T, Wilhite SE, Ledoux P, et al. NCBI GEO: archive for functional genomics data sets—update. *Nucleic Acids Res.* 2012;41(D1):D991-D995. doi:10.1093/nar/gks1193
53. Uhlen M, Zhang C, Lee S, et al. A pathology atlas of the human cancer transcriptome. *Science.* 2017;357(6352):2507. doi:10.1126/science.aan2507